Quantum Automation

Q-Elite Platform

Research & Development Followed By Extreme Automation

01

COMPANY

Involved In research, development followed by extreme Automation

02

TARGET

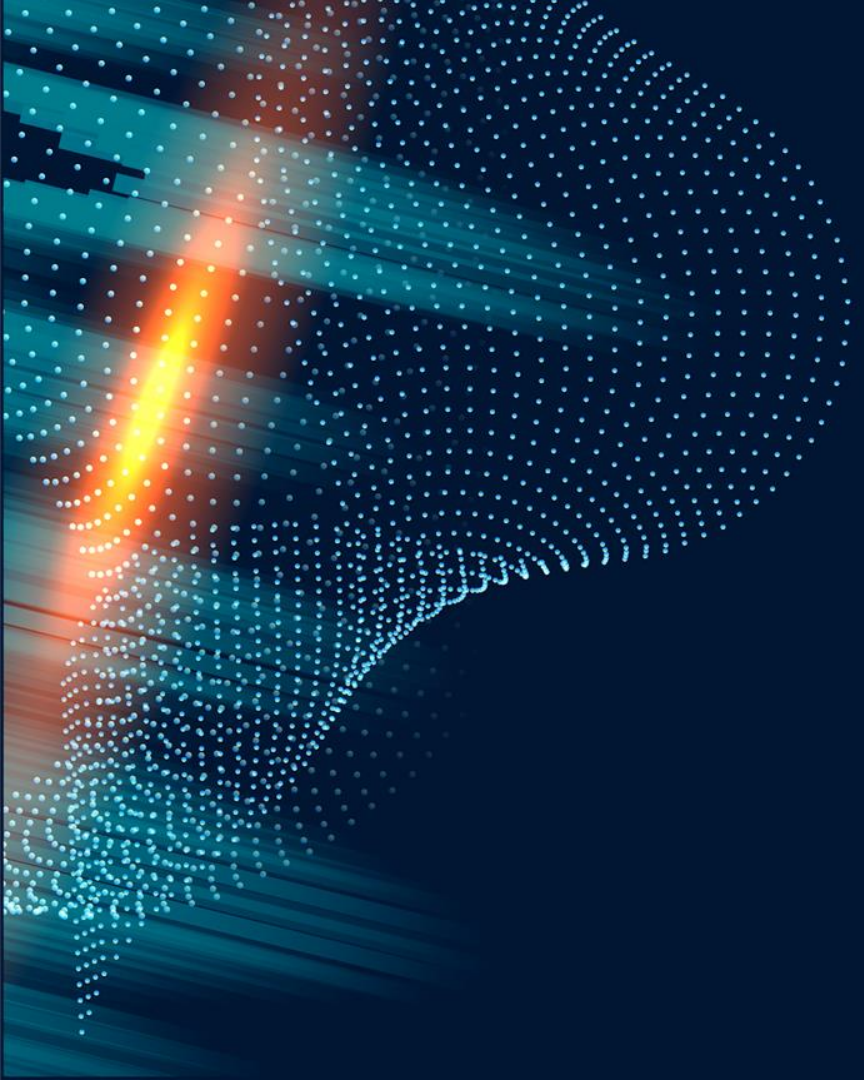
Develop AI/ML, Quantum Computing approach to implement secure ,intelligent Automation

03

SERVICES

End-to-End Automation, AI/ML based, zero touch provisioning

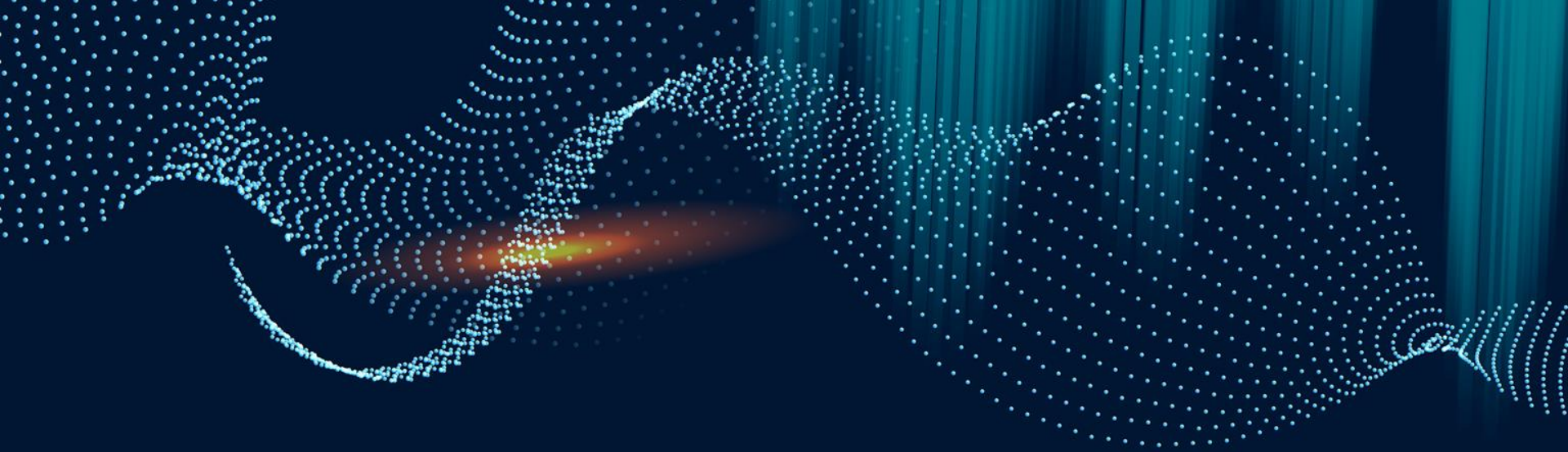
Automating enterprise networks, cloud, and data center operations.



Q-Elite Platform

Autonomous Infrastructure Operations

Quantum Automation provides End-To-End and extreme Automation solution to replace slow, outdated manual processes with streamlined, automated workflows, leading to increased efficiency and accuracy...



Key Feature’s

Quantum Automation

Q-Elite Platform

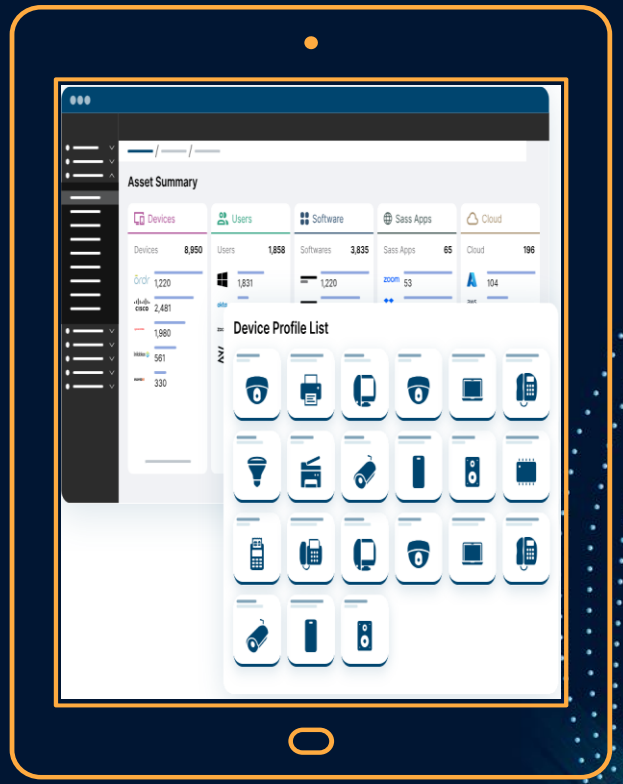
Device Auto Discovery



Device Auto Discovery feature discovers all the devices along with connectivity topology

Device Inventory Discovery

Device Inventory feature list the device's along with it's IP, type, SNCP details

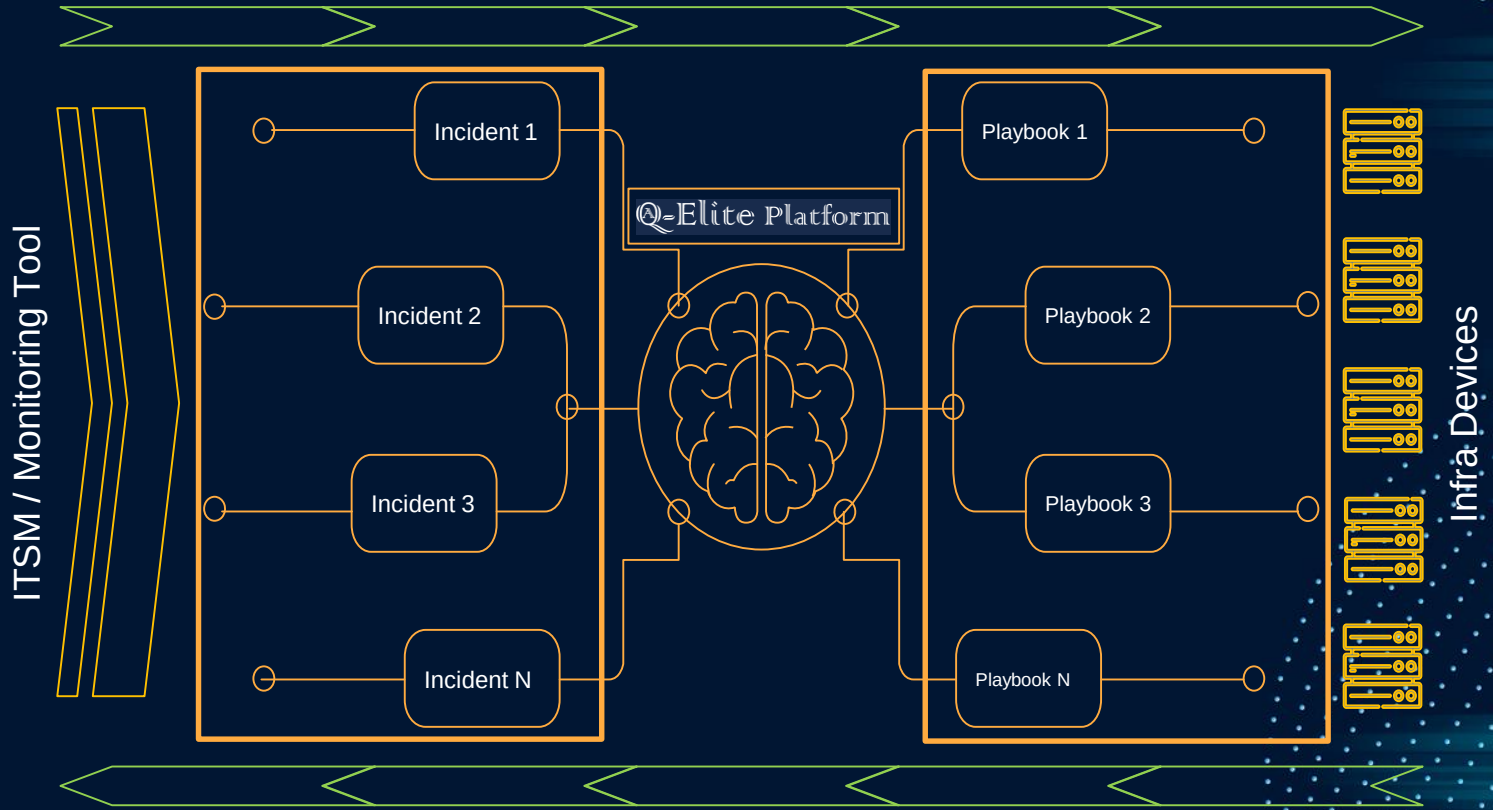


Device Reachability

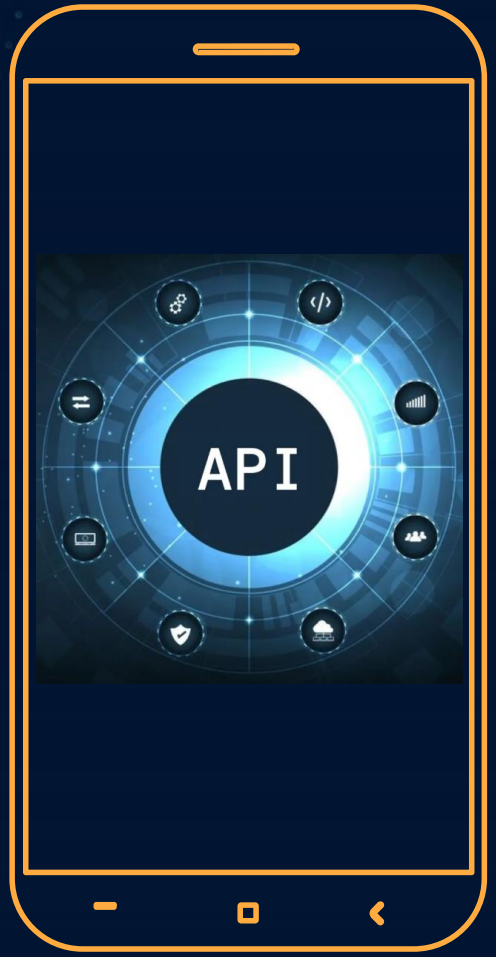


Device Reachability
Feature discovers all the infrastructure devices along with reachability

Auto Heal & End-To-End Automation Capabilities



API Based Automation



API based Automation feature via CURL command provide desired output with no need of Playbook development

RealTime CPU Utilization/Monitoring



RealTime CPU Utilization feature monitors the VM, Device's, Cloud processor real time utilization as per defined interval span

Vulnerability Discovery

Admin Panel

Q-Elite Platform | Quantum Automation | admin | Logout

Auto Discovery | Job Scheduling | Device Inventory | Device Reachability | Auto Heal | API Autodiscovery | Playbook Repository | Platform API | API Automation | Terminal | More

Vulnerability Scanner

Check all devices in an IP range for known vulnerabilities.

192.168.1.4 | 192.168.1.6 | Scan | Export CSV | Dark Mode

Total Devices: 3 | Vulnerable Devices: 2 | Total Vulnerabilities: 106 | Vendors: unknown,

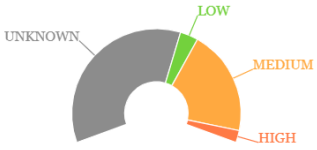
192.168.1.4
unknown
unknown
Status: Scan complete

192.168.1.5
Status: No SNMP response

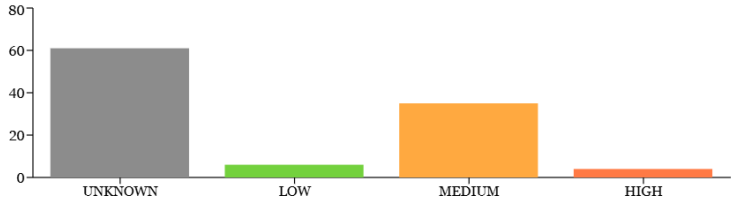
192.168.1.6
unknown
unknown
Status: Scan complete

Severity Distribution

Tilted 3D



Severity Count

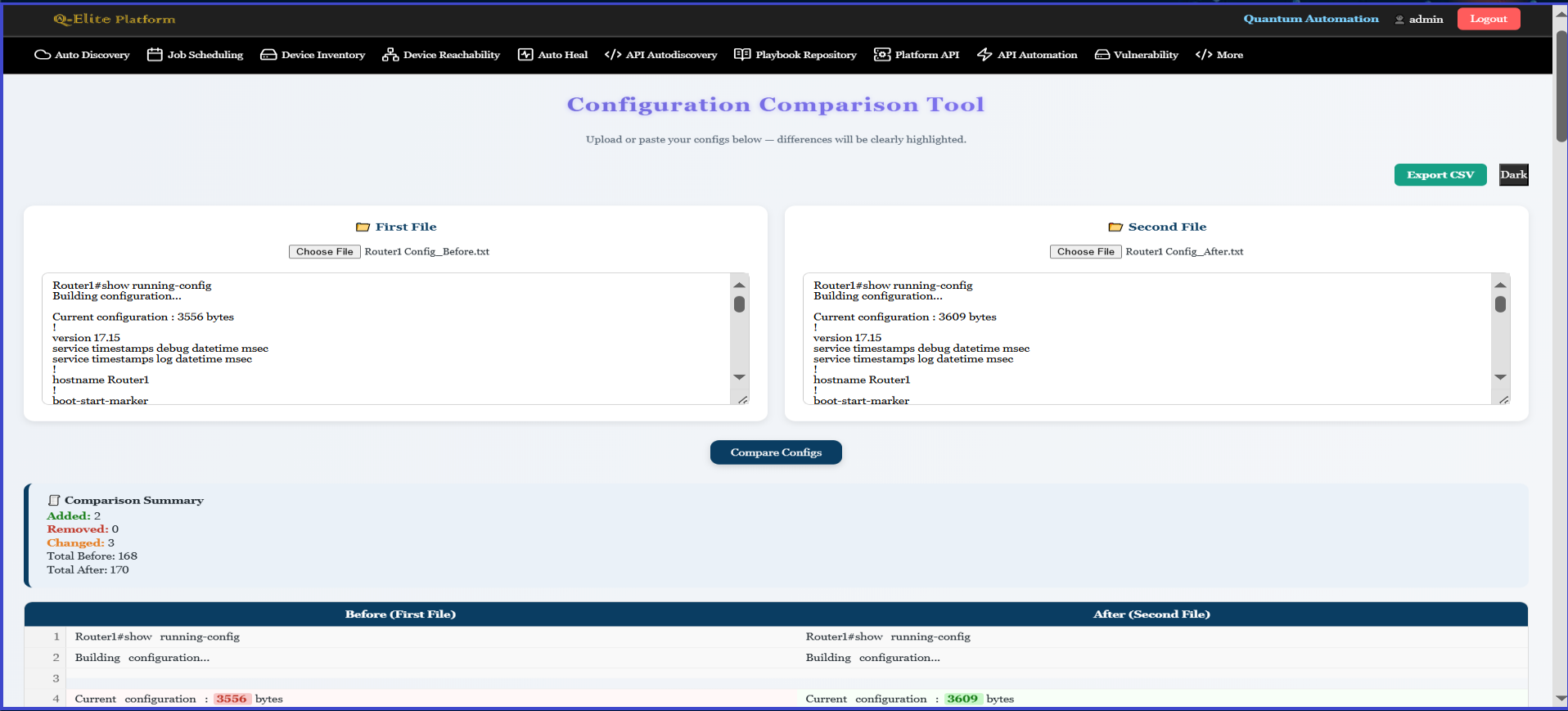


Vulnerabilities for 192.168.1.4

IP	Vendor	Product	Version	CVE ID	Severity	Score	Description
192.168.1.4	unknown	unknown	6.3	CVE-2005-4694	UNKNOWN	N/A	Unspecified vulnerability in the www_add method in Asset.pm in Plain Black WebGUI 6.3.0 and other versions before 6.7.6 ...
192.168.1.4	unknown	unknown	6.3	CVE-2006-5974	UNKNOWN	N/A	fetchmail 6.3.5 and 6.3.6 before 6.3.6-rc4, when refusing a message delivered via the mda option, allows remote attacker...

Configuration Comparison

- A feature to compare the configuration of the device to identify differences.
- This typically highlight changes in a side-by-side format, using color-coding for additions(green), deletions(red), and modifications(orange), to help identify unauthorized changes, troubleshoot issues, or verify that configurations are consistent.



Link and Bandwidth Utilization, Latency, Jitter & Wander

Admin Panel

Q-Elite Platform | Quantum Automation | admin | Logout

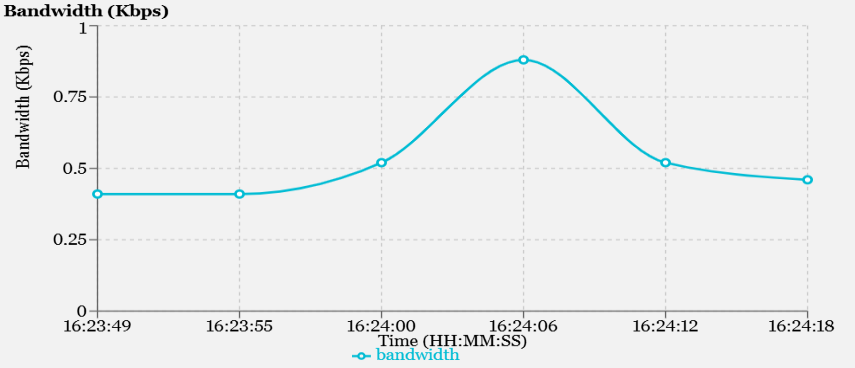
Auto Discovery | Job Scheduling | Device Inventory | Device Reachability | Auto Heal | API Autodiscovery | Playbook Repository | Platform API | API Automation | Vulnerability | More

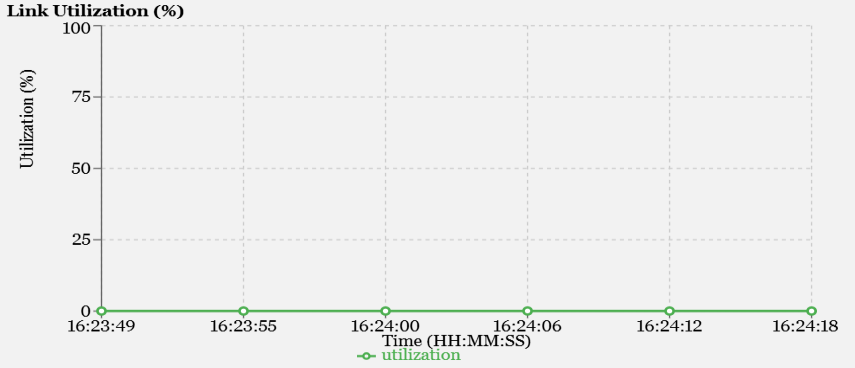
Link Utilization Monitor

Dark Mode

IP Address: 192.168.126.111 | Interval (sec): 2 | Bandwidth Unit: Kbps

▶ Start Monitoring | ■ Stop





Latency (ms)	Jitter (ms)	Wander (ms)
16.42	31.03	0

Observability Trace Feature

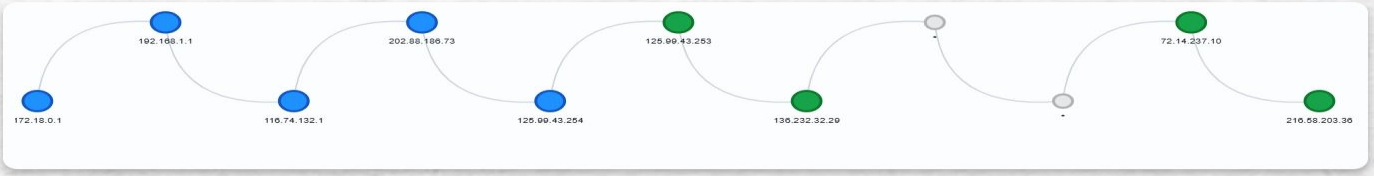
Q-Elite Platform | Quantum Automation | admin | Logout

Auto Discovery | Job Scheduling | Device Inventory | Device Reachability | Auto Heal | API Autodiscovery | Playbook Repository | Platform API | API Automation | Vulnerability | More

Observability Trace

Observability Graph

Fit



Local | Remote

Destination

www.google.com

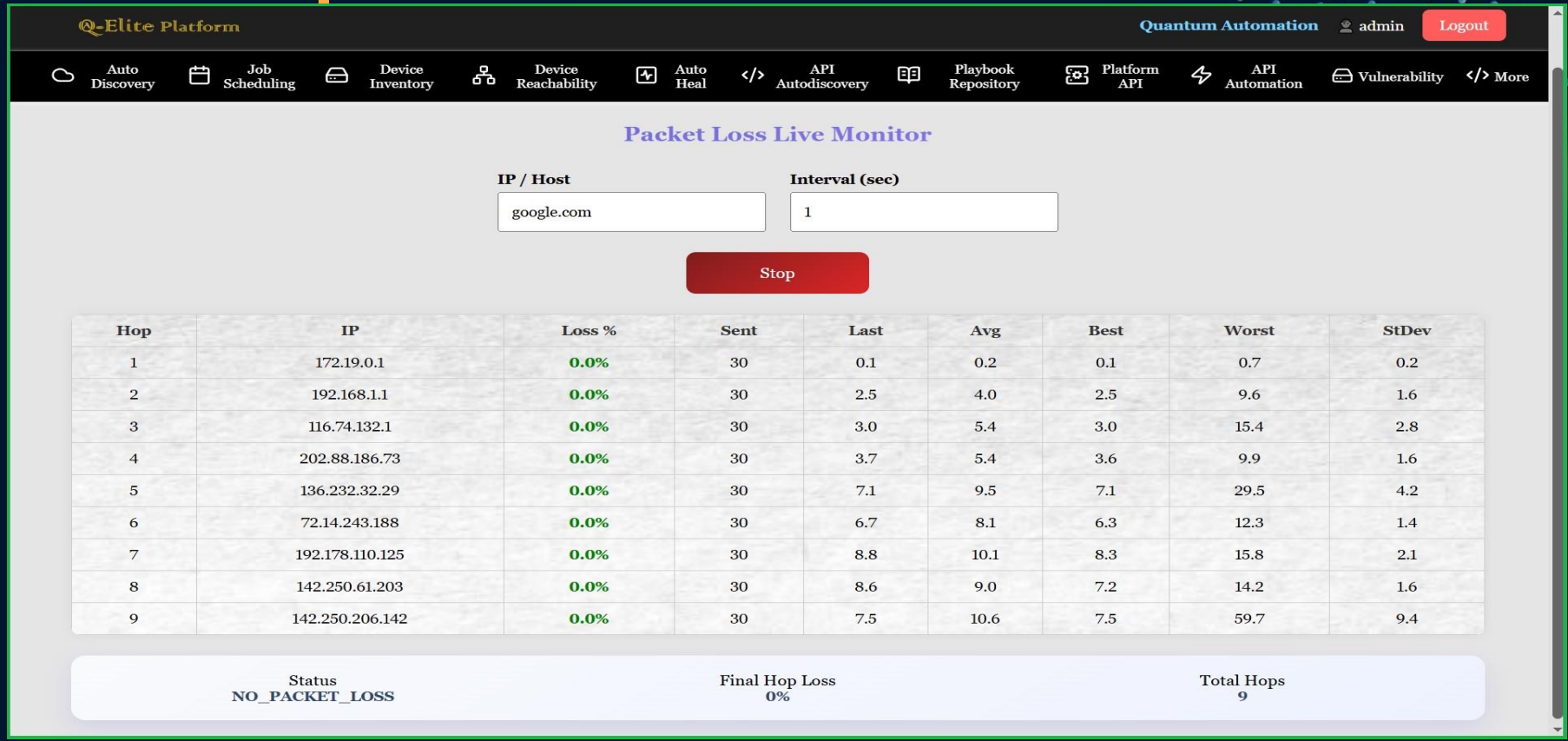
Run Local

IP Detected: 216.58.203.36

Parsed Hops

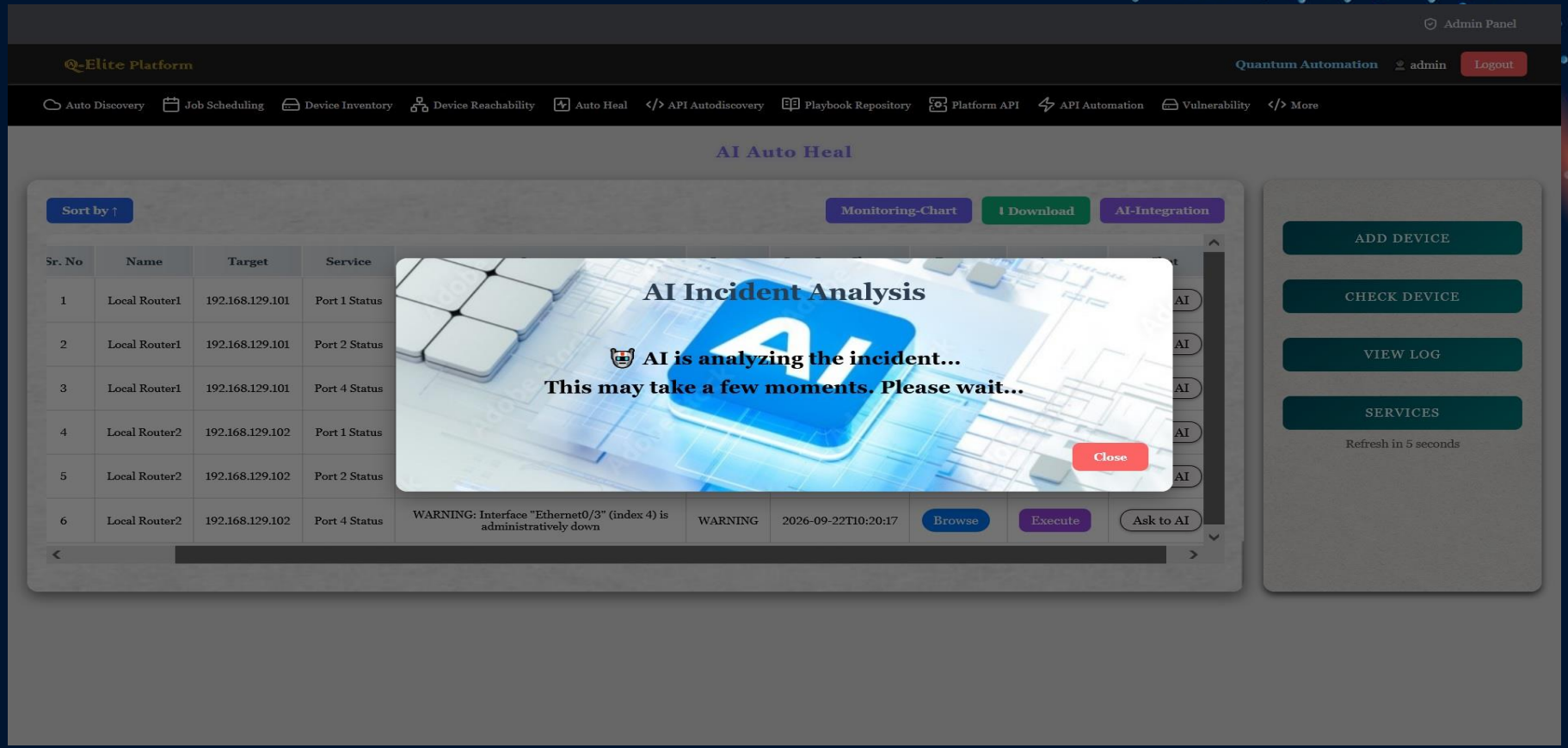
#	HOST / IP	AVG (MS)	STATUS	DEVICE
1	172.18.0.1	2.33	ok	router
2	192.168.1.1	2.07	ok	router
3	116.74.132.1	3.24	ok	router
4	202.88.186.73	4.53	ok	router
5	125.99.43.254	3.91	ok	router
6	125.99.43.253	4.49	ok	switch
7	136.232.32.29	21.91	ok	switch
8	*	—	timeout	unknown
9	*	—	timeout	unknown
10	72.14.237.10	11.36	ok	switch
11	216.58.203.36	11.48	ok	switch

Packet Capture Feature



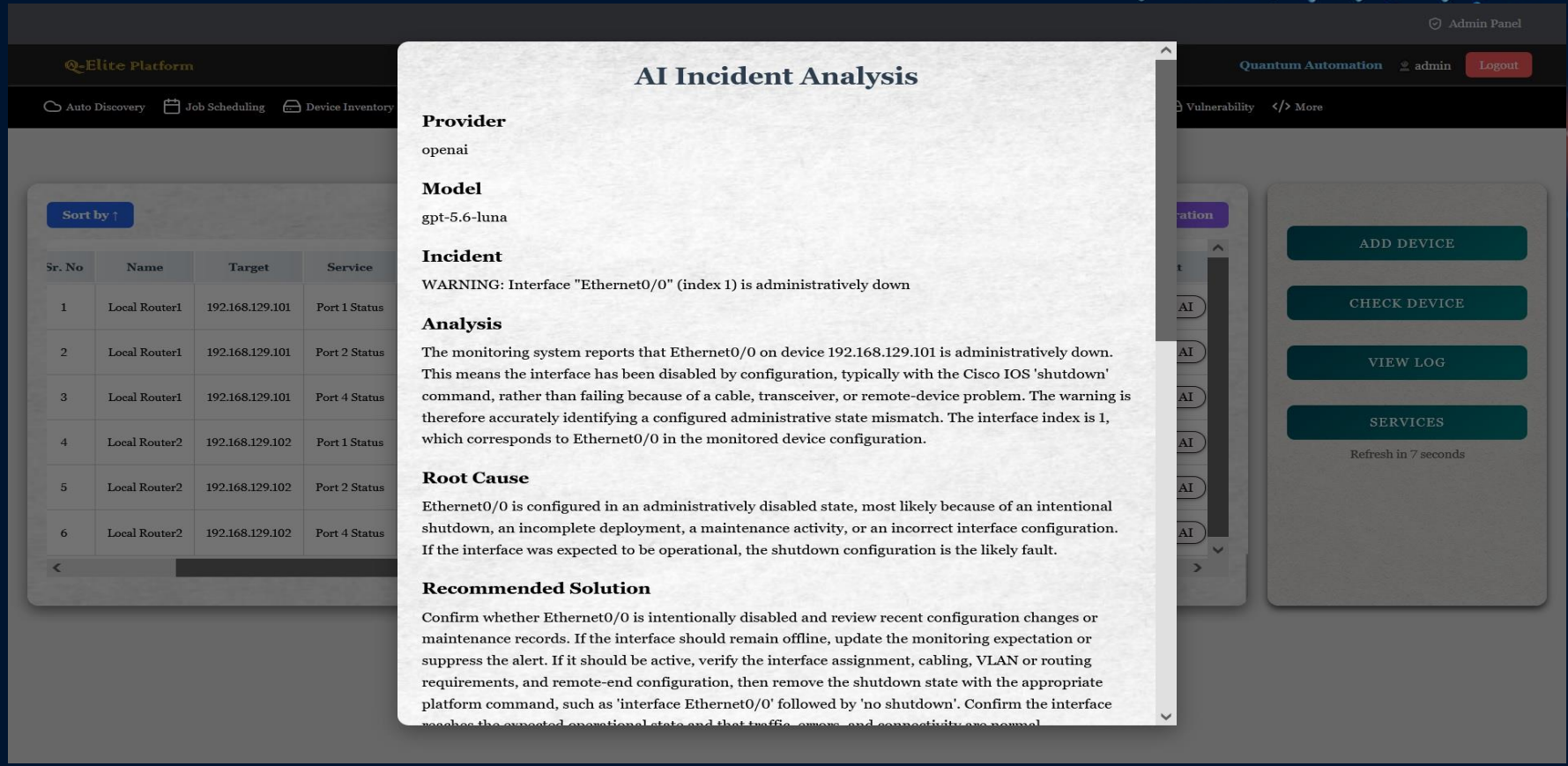
- Packet loss measures the percentage of network packets that fail to return from each node (hop) along a path, combining traceroute and ping for real-time analysis
- Packet loss below 5% is usually considered acceptable, with 10% not usually noticeable; however, 18-20% suggests significant congestion.

Device Monitoring



- Network device monitoring is the continuous observation of infrastructure components—like Routers, Switches, Firewalls, Servers, Load Balancer devices as well Website, VM's to track availability, health, and performance

AI-Auto Heal and Device Monitoring



- AI-Auto Heal and Monitoring feature is breakthrough in incident Analysis, Root Cause, Recommended Solution followed-by auto heal playbook via AI Model of individual choice out of four LLM/AI Model

Automation Platform Executable !!!

Infrastructure Automation levereges any of the 9 following playbook scripts

- Ansible (.yml, .yaml)
- Python (.py)
- Ruby (.rb)
- Perl (.pl)
- BASH (.sh)
- Shell Scripting (.sh)
- CHEF (.rb)
- Lua (.lua)
- Groovy (.groovy)
- Terraform (.tf)
- Java (.java)
- Puppet (.pp)
- SaltStack (.sls)

AUTOMATION ACROSS Data Center Technologies !!!



Why Quantum Automation Platform ???

- Device Auto Discovery feature via SNMP as well via API
- Device inventory discovery
- Scheduling of Automation task
- Auto heal feature with zero touch incident healing capabilities
- Platform capable of executing Playbook written in any out of the 9 Automation script
- API based Automation capabilities
- Containerized zero touch deployment with download, install, setup timeframe is within 5 - 10min
- Platform cost is competitive compare to similar solutions

We Believe in Research & Development Followed By Automation



Use Case List

- New Device On-Boarding (Cisco, Juniper, HP Aruba)
- Device OS upgrade (Routers, Switch, Firewall)
- Zero Touch Provisioning
- VLAN/Port Configuration
- Device Backup (Scheduling)
- Device Health-Check (Periodic Scheduling)
- Device Configuration Comparison
- Creation of Node, Pool & Virtual IP in F5 Load Balancer
- DNS Record Creation/Modification
- Firewall ACL & NAT creation
- Guest Wi-Fi Access
- URL Blacklisting and Whitelisting in Firewall

A TIMELINE ALWAYS WORKS WELL



Quantum Automation Approach Towards Automation Scope

Proactive Step By Step Approach

- Understanding pain points from client existing Infrastructure environment along with multi vendor challenges
- Carrying out due diligence of existing environment and understanding manual process execution flow
- Detail study carrying manual efforts vs Automation approach with time ,FTE hour, error free zero touch execution
- Based on detail study of client environment recommend best Automation tool along with comparison
- Targeting/Identifying top 10 use case's to implement via Automation achieving 15-30% FTE effort saving
- Defining tools deployment , implementation & execution timeline along with finalised use case timeline

Infrastructure Automation Capabilities !!!



Infrastructure Virtualization

Infrastructure topology discovery and visualization



Data Analytics

Data gathering and analyse ,event correlation



Task Automation

SaaS as well on-prem Infrastructure Automation



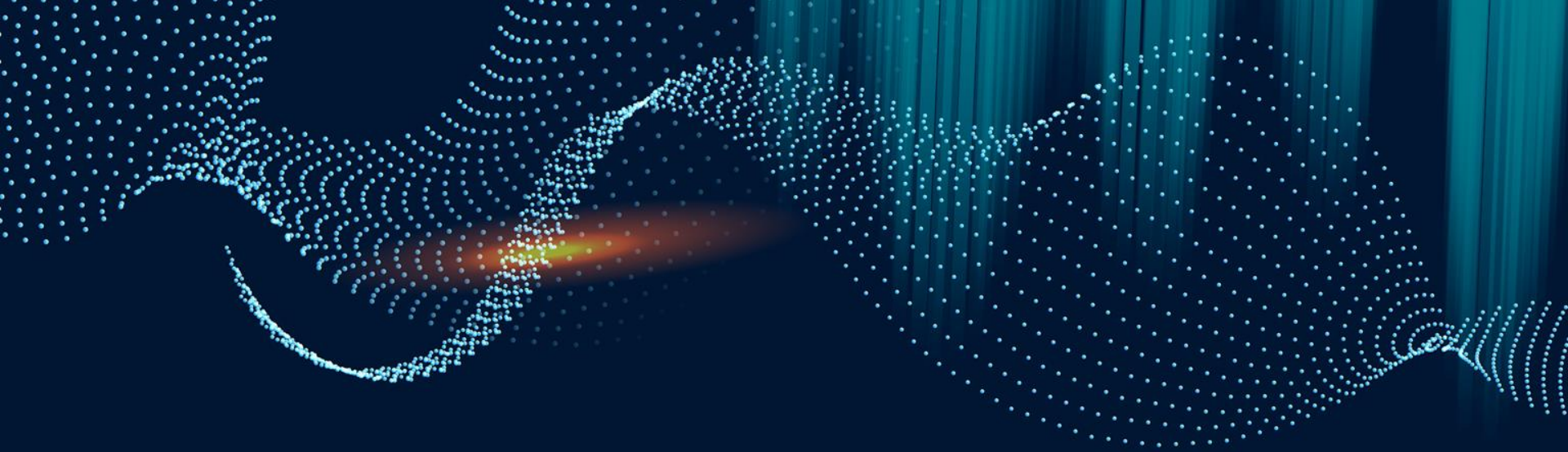
Extreme Automation

The convergence of AI/ML, observability and cloud computing elevate the Automation, enabling client business to grow and thrive with ease

Let’s Explore Automation

To deliver better Automation experiences, Quantum Automation innovation capabilities empower client Automation journeys to drive business growth.





A GOOD WAY TO BEGIN THE AUTOMATION JOURNEY

Human error free ,end to end zero touch, FTE savings, L1 utilisation instead L3, Auto heal Automation capabilities

AUTOMATION SUMMARY !!!

Explore Automation as much as possible ...

- Zero Touch provisioning
- End-To-End Automation capabilities
- Human error free
- Infrastructure Observability
- Junior resource(L1) capable to handle SME(L3) task via Automation
- 95% above efforts saving with 30% overall resource optmization possible
- 30% above straight financial savings
- API Integration advantage
- Go Green carbon neutral

Our Infrastructure Automation Solution Partners & Vendors...

SolarWinds	Observability, Network/System/Database/IT Service Management,
AlgoSec	Firewall Auditing/Compliance/management, Application Discovery, Incident reporting
Gluware	Brownfield Automation, Intent-Based networking, multi vendor platform
Netbrain	No Code Automation Platform, Cloud/Network/Security Automation solution
Nagios	Infrastructure monitoring, enhance network reliability and security
AWS & Google cloud	Official channel partner of Amazon AWS and Google Cloud service provider

- **Juniper**
- **HPE Aruba**
- **Cisco**
- **Extreme Networks**
- **Palo Alto**
- **Arista**

Platform License term & Setup Specifications

License Term	Year	RAM	Processor	Core	Hard Disk
1 Admin + 9 Members	1	8GB	4	1	200GB to 250GB
1 Admin + 24 Members	3	16GB	6	2	250GB to 300GB
1 Admin + 49 Members	5	24GB	8	2	300GB to 400GB

Setup	Hardware/Software
1 Linux VM	Ubuntu/RHEL/CentOS 64-bit
Number of Processors	4
Number of Cores per processor	1-4
Firewall/Network Ports to open	HTTP: 80 , HTTPS: 443 & SSH-22
VM Pre-installed	Docker

THANKS!

Do you have any questions?
connect@quantuma.ai

(+91) 91723 01688
www.quantuma.ai

QuantumA®

Q-Elite Platform

